

Closed-Form Richardson-Number Stability Functions from Monin–Obukhov Similarity Theory: Re-Derivation, Generalization, and Implementation Notes

A Technical Working Reference for Boundary-Layer Modeling

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Revising and Extending England & McNider (1995)

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Abstract

This document provides a mathematically rigorous, fully self-consistent reference for transforming Monin–Obukhov Similarity Theory (MOST) flux-profile functions $\phi_m(\zeta)$ and $\phi_h(\zeta)$ into explicit, closed-form gradient Richardson number (Ri) stability functions $f_m(Ri)$ and $f_h(Ri)$. We document and correct a historical transcription error in the $\zeta \leftrightarrow Ri$ mapping relation found in early literature, highlight the general master equation governing the stable boundary layer (SBL) for arbitrary heat profile exponents α_h , detail the step-by-step derivation of the unstable branch cubic equation, derive near-neutral series expansions via series reversion, and provide an explicit integration of surface roughness z_0 for discrete boundary-layer models.

1 Notation & Nomenclature

To reduce cognitive load and prevent ambiguity across local point-gradient theory, bulk atmospheric grid layers, and numerical implementations, the notation defined in Table 1 is maintained throughout this memorandum.

2 Theoretical Foundations & Coupling Identities

In first-order eddy-diffusivity closures (K -theory), turbulent exchange coefficients for momentum (K_m) and heat (K_h) are parameterized using mixing-length theory driven by mean vertical shear $S = |\partial u / \partial z|$:

$$K_m = l_m^2 S f_m(Ri), \quad K_h = l_h^2 S f_h(Ri) \quad (1)$$

Table 1: Summary of Mathematical Symbols and Definitions

Symbol	Description	Definition / Relation
Ri_g	Gradient Richardson Number	$\frac{g}{\theta_0} \frac{\partial \theta / \partial z}{(\partial u / \partial z)^2} = \zeta \frac{\phi_h(\zeta)}{\phi_m^2(\zeta)}$
Ri_b	Bulk Layer Richardson Number	$\frac{g}{\theta_0} \frac{\Delta \theta \Delta z}{(\Delta u)^2}$
Ri_c	Critical Richardson Number	$\lim_{\zeta \rightarrow \infty} Ri_g(\zeta) = \frac{1}{\beta_m}$ (Limiting value for turbulence)
Ri	Generic Local Richardson Number	Used in general algebraic derivations ($Ri \equiv Ri_g$)
ζ	Dimensionless Monin–Obukhov Stability	z/L
L	Monin–Obukhov Length Scale	$\frac{u_*^2 \theta_0}{\kappa g \theta_*}$
f_m, f_h	Ri -Based Stability Correction Functions	$K_m = l_m^2 S f_m(Ri), \quad K_h = l_h^2 S f_h(Ri)$
ϕ_m, ϕ_h	MOST Non-Dimensional Profile Functions	$\frac{\kappa z}{u_*} \frac{\partial u}{\partial z}, \quad \frac{\kappa z}{\theta_*} \frac{\partial \theta}{\partial z}$

where $l_m = l_h = \kappa z$ near the surface. Conversely, Monin–Obukhov Similarity Theory (MOST; Businger et al. 1971; Dyer 1974) defines diffusivities as:

$$K_m = \frac{\kappa z u_*}{\phi_m(\zeta)}, \quad K_h = \frac{\kappa z u_*}{\phi_h(\zeta)} \quad (2)$$

Equating K -theory with MOST using the surface-layer profile relation $S = \frac{u_*}{\kappa z} \phi_m(\zeta)$ yields the fundamental coupling definitions:

$$f_m(Ri) = \frac{1}{\phi_m^2(\zeta)} \quad (3)$$

$$f_h(Ri) = \frac{1}{\phi_m(\zeta) \phi_h(\zeta)} = \frac{f_m^{1/2}(Ri)}{\phi_h(\zeta)} \quad (4)$$

2.1 Correction of the Historical $\zeta \leftrightarrow Ri$ Transformation

The gradient Richardson number Ri_g is defined via MOST profiles as:

$$Ri = \frac{g}{\theta_0} \frac{\frac{\partial \theta}{\partial z}}{\left(\frac{\partial u}{\partial z}\right)^2} = \zeta \frac{\phi_h(\zeta)}{\phi_m^2(\zeta)} \quad (5)$$

Substituting (3) and (4) into (5) yields the exact relation connecting Ri and ζ :

$$\phi_m(\zeta) = f_m^{-1/2}(Ri), \quad \phi_h(\zeta) = \frac{f_m^{1/2}(Ri)}{f_h(Ri)} \quad (6)$$

Inserting these back into (5):

$$Ri = \zeta \frac{f_m^{1/2}/f_h}{f_m^{-1}} = \zeta \frac{f_m^{3/2}(Ri)}{f_h(Ri)} \implies \boxed{\zeta = Ri \frac{f_h(Ri)}{f_m^{3/2}(Ri)}} \quad (7)$$

Historical Note on Equation (7): In earlier literature—most notably Equation (17) of England & McNider (1995)—the exponent on f_m in the denominator was misprinted as 1 rather than $3/2$ (i.e., $\zeta = Ri \frac{f_m}{f_h}$). While $f_m/f_h = 1$ when $Pr_t = 1$, for general turbulent Prandtl numbers ($Pr_t \neq 1$), the $f_m^{3/2}$ power is mathematically required to preserve exact internal consistency with downstream derivations.

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3 Stable Boundary Layer Formulation ($\zeta \geq 0$)

We generalize the stable non-dimensional profiles by retaining a linear momentum function ($\alpha_m = 1$) while allowing a general exponent α_h for heat:

$$\phi_m(\zeta) = 1 + \beta_m \zeta, \quad \phi_h(\zeta) = (1 + \beta_h \zeta)^{\alpha_h} \quad (\zeta \geq 0) \quad (8)$$

Defining $x \equiv f_m^{1/2}(Ri) = \phi_m^{-1}(\zeta)$, we invert $\phi_m(\zeta)$ to obtain:

$$\zeta = \frac{1-x}{\beta_m x} \quad (9)$$

3.1 Central Master Equation for $x \equiv f_m^{1/2}$

Substituting (9) into (7) gives the explicit expression for $f_h(Ri)$:

$$f_h(Ri) = \frac{x^2(1-x)}{\beta_m Ri} \quad (10)$$

Equating $\phi_h = x/f_h = \frac{\beta_m Ri}{x(1-x)}$ with $\phi_h(\zeta) = \left(1 + \beta_h \frac{1-x}{\beta_m x}\right)^{\alpha_h}$ produces the ****central master equation**** governing all stable FIRST-order closures:

$$\boxed{\left[\frac{Ri \beta_m}{x(1-x)}\right]^{1/\alpha_h} = \frac{\beta_m x + \beta_h(1-x)}{\beta_m x}} \quad (11)$$

3.2 Case 1: Standard Linear Heat Profile ($\alpha_h = 1$)

Setting $\alpha_h = 1$ reduces (11) directly to a quadratic polynomial in x :

$$Ri(\beta_h - \beta_m)x^2 - (1 + Ri \beta_h)x + 1 = 0 \quad (12)$$

Taking the physically meaningful root ($x \rightarrow 1$ as $Ri \rightarrow 0$) yields the exact closed-form momentum stability function:

$$f_m(Ri) = x^2 = \left[\frac{(1 + Ri \beta_h) - \sqrt{(1 + Ri \beta_h)^2 - 4Ri(\beta_h - \beta_m)}}{2Ri(\beta_h - \beta_m)} \right]^2 \quad (13)$$

The corresponding heat function $f_h(Ri)$ is evaluated directly via (10).

3.3 Case 2: Businger–Dyer Quadratic Heat Profile ($\alpha_h = 2$)

Setting $\alpha_h = 2$ and squaring both sides of (11) clears fractional exponents, yielding a cubic polynomial $a_3x^3 + a_2x^2 + a_1x + a_0 = 0$:

$$\begin{aligned} a_3 &= -(\beta_m - \beta_h)^2 \\ a_2 &= (\beta_m - \beta_h)^2 - 2\beta_h(\beta_m - \beta_h) \\ a_1 &= 2\beta_h(\beta_m - \beta_h) - \beta_h^2 - Ri \beta_m^3 \\ a_0 &= \beta_h^2 \end{aligned} \quad (14)$$

3.4 Equal-Parameter Limit ($\beta_m = \beta_h = \beta$)

When $\beta_m = \beta_h = \beta$, $a_3 = 0$ and $a_2 = 0$. The quadratic (12) degenerates to a simple linear equation in x :

$$(1 + \beta Ri)x = 1 \implies x = \frac{1}{1 + \beta Ri} \quad (15)$$

which yields:

$$f_m(Ri) = (1 + \beta Ri)^{-2}, \quad f_h(Ri) = (1 + \beta Ri)^{-3} \quad (16)$$

Distinction Note: This result demonstrates that quadratic closures like $(1 - \beta Ri)^2$ stem from distinct physical assumption sets ($Pr_t = 1$ toy models) rather than standard linear MOST profile inversions.

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4 Unstable Boundary Layer Formulation ($\zeta < 0$)

For unstable stratification ($\zeta < 0$), standard Businger–Dyer profile functions take the generalized exponential form:

$$\phi_m(\zeta) = (1 - \beta_m \zeta)^{-\alpha_m}, \quad \phi_h(\zeta) = (1 - \beta_h \zeta)^{-\alpha_h} \quad (17)$$

For classical coefficients ($\alpha_m = 1/4$, $\alpha_h = 1/2$), the stability functions map in terms of ζ as:

$$f_m(\zeta) = (1 - \beta_m \zeta)^{1/2} \quad (18)$$

$$f_h(\zeta) = (1 - \beta_h \zeta)^{1/2} (1 - \beta_m \zeta)^{1/4} \quad (19)$$

4.1 Master Cubic Equation in $x \equiv f_m^2$

Expressing f_h directly in terms of f_m :

$$f_h(f_m) = f_m^{1/2} \left[1 - \frac{\beta_h}{\beta_m} (1 - f_m^2) \right]^{1/2} \quad (20)$$

Substituting (20) into the corrected transformation identity $\zeta^2 = Ri^2 \frac{f_h^2}{f_m^3}$ and defining $x \equiv f_m^2(Ri)$ yields the exact master cubic equation for unstable conditions (see Appendix A for full intermediate steps):

$$x^3 - 2x^2 + \left(1 - Ri^2 \beta_m \beta_h \right) x - Ri^2 \beta_m (\beta_m - \beta_h) = 0 \quad (21)$$

4.2 Analytical Solution for Equal Coefficients ($\beta_m = \beta_h = 16$)

When $\beta_m = \beta_h = 16$, the constant term in (21) vanishes. Factoring out $x \neq 0$:

$$x^2 - 2x + (1 - 256Ri^2) = 0 \implies (1 - x)^2 = 256Ri^2 \implies x = 1 - 16Ri \quad (22)$$

Recalling $x \equiv f_m^2$, we recover the analytical expression:

$$f_m(Ri) = (1 - 16Ri)^{1/2} \quad (23)$$

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5 Series Reversions & Pedagogical Models

5.1 Near-Neutral Series Expansions via Series Reversion

When exact inversion is mathematically intractable, Taylor expanding $\phi_m(\zeta) = 1 + \beta_m \zeta + \gamma_m \zeta^2$ and $\phi_h(\zeta) = 1 + \beta_h \zeta + \gamma_h \zeta^2$ around $\zeta = 0$ provides a forward series for $Ri(\zeta)$:

$$Ri_g(\zeta) = \zeta + (\beta_h - 2\beta_m)\zeta^2 + (3\beta_m^2 - 2\beta_m\beta_h - 2\gamma_m + \gamma_h)\zeta^3 + O(\zeta^4) \quad (24)$$

Reverting this series yields the inverted expression for $\zeta(Ri_g)$:

$$\zeta(Ri_g) = Ri_g + (2\beta_m - \beta_h)Ri_g^2 + (5\beta_m^2 - 6\beta_m\beta_h + 2\beta_h^2)Ri_g^3 + O(Ri_g^4) \quad (25)$$

Substituting $\zeta(Ri_g)$ into $f_m = \phi_m^{-2}$ and $f_h = (\phi_m \phi_h)^{-1}$ yields the low-order Ri -series forms:

$$f_m(Ri_g) = 1 - 2\beta_m Ri_g - \beta_m(\beta_m - 2\beta_h)Ri_g^2 + O(Ri_g^3) \quad (26)$$

$$f_h(Ri_g) = 1 - (\beta_m + \beta_h)Ri_g - (\beta_m^2 - 2\beta_h^2 + \gamma_m + \gamma_h)Ri_g^2 + O(Ri_g^3) \quad (27)$$

5.2 One-Parameter Toy Model with Finite Ri_c

Under the explicit assumption of equal profile parameters ($\beta_m = \beta_h = \beta \implies Pr_t = 1$), the transformation reduces to $Ri_g = \frac{\zeta}{1+\beta\zeta}$. Inverting directly gives:

$$\zeta(Ri_g) = \frac{Ri_g}{1 - \beta Ri_g} \implies \phi_m(Ri_g) = \frac{1}{1 - \beta Ri_g} \quad (28)$$

Defining the critical Richardson number $Ri_c \equiv 1/\beta$, the closure functions reduce to:

$$f_m(Ri_g) = f_h(Ri_g) = \left(1 - \frac{Ri_g}{Ri_c}\right)^2 = \left(\frac{Ri_c - Ri_g}{Ri_c}\right)^2 \quad (29)$$

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6 Surface Layer Roughness (z_0) Integration

In numerical mesoscale and boundary-layer models (e.g., Louis 1979; McNider & Pielke 1981; England & McNider 1995), the finite vertical grid spacing requires calculating fluxes across a discrete layer from surface roughness height z_0 to first model level z_1 . The bulk Richardson number across this layer is defined as:

$$Ri_b = \frac{g}{\theta_0} \frac{[\theta(z_1) - \theta(z_0)](z_1 - z_0)}{[u(z_1) - u(z_0)]^2} \quad (30)$$

Integrating continuous MOST profile equations $\frac{\partial u}{\partial z} = \frac{u_*}{\kappa z} \phi_m$ and $\frac{\partial \theta}{\partial z} = \frac{\theta_*}{\kappa z} \phi_h$ from z_0 to z_1 yields the integrated profiles:

$$u(z_1) = \frac{u_*}{\kappa} \left[\ln\left(\frac{z_1}{z_0}\right) - \Psi_m\left(\frac{z_1}{L}\right) \right], \quad \theta(z_1) - \theta(z_0) = \frac{\theta_*}{\kappa} \left[\ln\left(\frac{z_1}{z_0}\right) - \Psi_h\left(\frac{z_1}{L}\right) \right] \quad (31)$$

Substituting these integrated profiles into (30) (assuming $z_1 \gg z_0$ such that $z_1 - z_0 \approx z_1$) expresses Ri_b in terms of integrated stability functions Ψ_m and Ψ_h :

$$Ri_b = \left(\frac{z_1}{L}\right) \frac{\ln(z_1/z_0) - \Psi_h(z_1/L)}{[\ln(z_1/z_0) - \Psi_m(z_1/L)]^2} \quad (32)$$

To ensure exact mathematical equivalence between local point gradient functions $Ri_g = \zeta \frac{\phi_h}{\phi_m^2}$ and discrete layer computations Ri_b , an **effective depth** z_{eff} is defined such that $Ri_g(z_{\text{eff}}/L) = Ri_b$:

$$z_{\text{eff}} = z_1 \left[\frac{\ln(z_1/z_0) - \Psi_m(z_1/L)}{\ln(z_1/z_0) - \Psi_h(z_1/L)} \right] \quad (33)$$

This formulation allows local Ri -based stability closures $f_m(Ri)$ and $f_h(Ri)$ to be evaluated directly using Ri_b at height z_{eff} without iterative root-finding.

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A Complete Intermediate Derivation of the Unstable Cubic Equation

This appendix provides every intermediate algebraic step for deriving the unstable master cubic equation (21).

Step 1: Squaring the Corrected Transformation Identity. Starting from the corrected relation (7):

$$\zeta = Ri \frac{f_h(Ri)}{f_m^{3/2}(Ri)} \implies \zeta^2 = Ri^2 \frac{f_h^2(Ri)}{f_m^3(Ri)} \quad (A1)$$

Step 2: Expressing ζ in terms of f_m . For Businger–Dyer momentum profile $\phi_m = (1 - \beta_m \zeta)^{-1/4}$, equation (18) gives:

$$f_m = (1 - \beta_m \zeta)^{1/2} \implies f_m^2 = 1 - \beta_m \zeta \implies \zeta = \frac{1 - f_m^2}{\beta_m} \quad (A2)$$

Step 3: Expressing f_h^2 in terms of f_m . For heat profile $\phi_h = (1 - \beta_h \zeta)^{-1/2}$, equation (19) gives:

$$f_h = (1 - \beta_h \zeta)^{1/2} (1 - \beta_m \zeta)^{1/4} = (1 - \beta_h \zeta)^{1/2} f_m^{1/2} \quad (A3)$$

Substituting $\zeta = \frac{1 - f_m^2}{\beta_m}$ into $(1 - \beta_h \zeta)$:

$$1 - \beta_h \zeta = 1 - \frac{\beta_h}{\beta_m} (1 - f_m^2) = \frac{\beta_m - \beta_h (1 - f_m^2)}{\beta_m} \quad (A4)$$

Squaring f_h :

$$f_h^2 = f_m \left[\frac{\beta_m - \beta_h + \beta_h f_m^2}{\beta_m} \right] \quad (A5)$$

Step 4: Substituting into Transformation Identity. Substituting ζ^2 from (A2) and f_h^2 from (A5) into (A1):

$$\left(\frac{1 - f_m^2}{\beta_m} \right)^2 = Ri^2 \frac{f_m \left(\frac{\beta_m - \beta_h + \beta_h f_m^2}{\beta_m} \right)}{f_m^3} \quad (A6)$$

Simplifying powers of f_m and β_m :

$$\frac{(1 - f_m^2)^2}{\beta_m^2} = \frac{Ri^2}{\beta_m f_m^2} [(\beta_m - \beta_h) + \beta_h f_m^2] \quad (A7)$$

Step 5: Change of Variable $x \equiv f_m^2$. Setting $x = f_m^2$:

$$\frac{(1 - x)^2}{\beta_m^2} = \frac{Ri^2}{\beta_m x} [(\beta_m - \beta_h) + \beta_h x] \quad (A8)$$

Multiplying both sides by $\beta_m^2 x$:

$$x(1 - x)^2 = Ri^2 \beta_m [(\beta_m - \beta_h) + \beta_h x] \quad (A9)$$

Expanding the left-hand side:

$$x(1 - 2x + x^2) = x^3 - 2x^2 + x \quad (\text{A10})$$

Expanding the right-hand side and grouping terms:

$$x^3 - 2x^2 + x = Ri^2 \beta_m (\beta_m - \beta_h) + Ri^2 \beta_m \beta_h x \quad (\text{A11})$$

Bringing all terms to the left-hand side yields equation (21):

$$x^3 - 2x^2 + (1 - Ri^2 \beta_m \beta_h)x - Ri^2 \beta_m (\beta_m - \beta_h) = 0 \quad (\text{A12})$$

This completes the formal proof.

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